Customer wants to use an existing fluorescence spectrometer to report glucose concentration and viable biomass directly during fermentation, justified only by fluorescence sensitivity, with no identified fluorescent target.

glucose | viable_biomass

**Over all cases (minimum per method):** NIR 21 · MIR 20 · Raman 20 · Fluorescence 18 · UV-Vis 18

DECISION STATUS **Provisional feasibility assessment**

### Raman leads (23/36); provisional until 'optical properties fluorescence' is checked.

**Raman, Raman probe** · 23/36

- Relative compatibility with the aqueous medium, not a ppm guarantee.
  `S01 p16` `REF D04` `R_Q1_R03`
- Broad target-signal link; no catalog entry = UNKNOWN, never exclusion.
  `S01 p4-5` `REF §3,5` `R_Q1_01`

**Client premise not supported:** High sensitivity alone does not show a direct signal of the target.
`R_Q1_F02` `S01 p16` `H03`

## Recommended methods

#1
**23** /36
range [6–32]

### Raman · Raman probe

Confidence: low | ⚠ critical question open

| | | | |
|---|---|---|---|
| Q1 | 7 /9 | Q1b | 4 /9 |
| Q2 | 5 /9 | Q3 | 7 /9 |
| Q4 | 2 /3 | | |

#2
**21** /36
range [4–32]

### NIR · diffuse reflection probe

Confidence: low | ⚠ critical question open

| | | | |
|---|---|---|---|
| Q1 | 4 /9 | Q1b | 4 /9 |
| Q2 | 6 /9 | Q3 | 7 /9 |
| Q4 | 0 /3 | | |

#3
**20** /36
range [4–32]

### MIR · ATR probe

Confidence: low | ⚠ critical question open

| | | | |
|---|---|---|---|
| Q1 | 4 /9 | Q1b | 4 /9 |
| Q2 | 5 /9 | Q3 | 7 /9 |
| Q4 | 0 /3 | | |

▼ More methods (2)

#4
**20** /36
range [4–32]

### UV-Vis · ATR probe

Confidence: low | ⚠ critical question open

| | | | |
|---|---|---|---|
| Q1 | 4 /9 | Q1b | 4 /9 |
| Q2 | 5 /9 | Q3 | 7 /9 |
| Q4 | 0 /3 | | |

#5
**18** /36
range [4–29]

## Fluorescence · fluorimetric probe

Confidence: low | ⚠ critical question open

| | | | |
|---|---|---|---|
| Q1 | 2 /9 | Q1b | 4 /9 |
| Q2 | 5 /9 | Q3 | 7 /9 |
| Q4 | 0 /3 | | |

## Questions for the customer

1. high `Sample · optical properties · fluorescence`

   **Does the matrix fluoresce (known fluorescent components, colour)?**

   ▼ Why

   if fluorescence=present: leader NIR

   `R_Q1_R02`

2. high `Sample · optical properties · opacity`

   **Is the sample clear, turbid or opaque at the measuring point, and does it change?**

   ▼ Why

   if opacity=clear: leader NIR

   `R_Q2_04`

3. high `Sample · sample state constraints · contact`

   **May a probe touch the product at the measuring point?**

   ▼ Why

   if contact=allowed: leader ['Raman|RAMAN_PROBE', 'MIR|MIR_ATR_PROBE', 'NIR|NIR_DIFFUSE_PROBE']; if contact=prohibited: leader Raman

   `R_Q2_03, R_Q2_06`

To improve accuracy, add the customer's answers to the text above and click Analyze again.

## Ruled out (1)

**UV_VIS · UV_LENS**
This interface collects light emitted by the sample itself; the sample is not a light source. `R_Q2_02`

## Decision path · factors of the leading candidate per stage, with source

## Q1

MATCH **+2** Broad target-signal link; no catalog entry = UNKNOWN, never exclusion.
`S01 p4-5` `REF §3,5` R_Q1_01

UNKNOWN **0** Separation of the target signal from the interferences of this matrix. MATCH only where an evidence row names the page (selectivity_source); no spectra in the materials = UNKNOWN, never a MISMATCH and never 'no interferences'. `S01 p5,10-11` R_Q1_04

UNKNOWN **0** Generic row: a known matrix background must not cover the target signal. Unknown overlap = UNKNOWN. Replaced by the specific rows R_Q1_R02 (Raman fluorescence) and R_Q1_M03 (MIR water). Soft only; a hard exclusion needs a proven overlap (R_Q1_02). `S01 p5,11` `REF §5` R_Q1_05

UNKNOWN **0** Independent validation material must be established, not assumed.
`REF D01/D02 experiment` `S01 p5,10` R_Q1_07

UNKNOWN **0** Unknown excitation/background = UNKNOWN; fluorescence without a spectrum does not forbid Raman. `S01 p11` `REF §5` R_Q1_R02

MATCH **+1** Relative compatibility with the aqueous medium, not a ppm guarantee. `S01 p16` `REF D04` R_Q1_R03

## Q1b

UNKNOWN **0** LOD/LOQ only from a scoped CapabilityProfile; the materials give none.
`spec 04 §1` `REF §3,5` R_Q1B_02

UNKNOWN **0** Achieved error vs allowed error needs a comparable metric; none in the materials.
`REF D01/D02 experiment` `spec 04` R_Q1B_04

UNKNOWN **0** A laboratory is not a known reference uncertainty. `REF D01,D02` `H01` R_Q1B_05

UNKNOWN **0** End-to-end latency needs a profile; spectral exposure alone is not the latency.
`spec 04 §1` R_Q1B_06

UNKNOWN **0** Both range ends must be inside the calibration/detector scope.
`REF D01,D02` `S01 p16,18` R_Q1B_08

UNKNOWN **0** Qualitative sensitivity rating: context only, never a score. `XLSX sensitivity + notes` R_Q1B_10

## Q2

MATCH **+1** Documented Method-interface edge (S01 Table 1). `XLSX` `S01 p4-6` R_Q2_01

UNKNOWN **0** A form missing from the list is UNKNOWN; MISMATCH only for a constructive conflict (solid vs liquid-only cell). `XLSX` `CL10` R_Q2_02

UNKNOWN **0** Excludes only this pair; other interfaces stay. Taking a sample out is R_Q2_10 (one cause, one rule). `S01 p5,10` R_Q2_03

UNKNOWN **0** A surface or local signal is not the bulk composition of a heterogeneous stream without a sampling plan. `S01 p8,13-14,17-18` `REF D08` R_Q2_07

UNKNOWN **0** Particle/bubble/surface variation must be covered by a sampling or averaging plan. Our recommendation to average is not a plan (no MATCH from it). `REF D01,D02` `S01 p15,17` R_Q2_08

### Q3

UNKNOWN **0** No numeric limit in the materials. Hot extrusion does not exclude MIR.
`S02 p14` `REF D07` R_Q3C_01

UNKNOWN **0** Fouling + contact interface needs a feasible cleaning/maintenance plan; a promise is not a match. `REF D02,D07` R_Q3C_04

MATCH **+2** Preference is soft; a lab interface stays valid for the feasibility experiment. `REF §3`
R_Q3C_05

UNKNOWN **0** Do not invent certification. `design comparator` R_Q3C_06

UNKNOWN **0** Long silica cables are documented; no universal allowed length.
`S02 p6-10,18` `S01 p15` R_Q3C_07

MATCH **+1** Generic family is a soft concept check; a real cable with proven mismatch is R_Q3P_03.
`S02 p6` `REF §5` R_Q3C_08

## Risks and assumptions

high Premise not supported: high fluorescence sensitivity alone does not show a direct signal of glucose or viable biomass. No fluorescent target is identified (none known), and the existing fluorescence spectrometer and any calibration cannot be assumed valid for these targets. A fluorescence signal of the broth could correlate with biomass by accident. What would still meet the goal: measure glucose in-line with Raman or NIR fiber probes after a lab test, and use the fluorescence instrument only as a possible indirect biomass or state indicator once it is shown against a viable-cell reference.

medium Ranking vs assessment: the ranking leads with Raman (23/36) for glucose, but for viable biomass NIR leads (21/36) and Raman is 20/36. All candidates are low confidence, the intervals are very wide (e.g. Raman [6, 32]), and the open fluorescence question could change the glucose leader to NIR. The ranking is a draft scoring on physics-level support, not on an AFS application. Scattering, which the card names as a typical biomass method, is excluded only because it is not in the method catalog, so a draft scoring rule outweighs physics here; note that it sees total particles, not viability.

high Viable biomass is not a direct spectral target for any listed method. Models may follow total cell mass, turbidity or glucose depletion, which are correlated in the process; designed samples (viable vs heat-killed cells) are needed to see this.

high Fluorescence background of broth or medium could swamp the Raman signal; this is unknown and is the critical unknown for the glucose lead. Check at the chosen laser wavelength; a longer wavelength reduces the problem.

medium Water bands and overlapping sugar bands can limit NIR sensitivity if glucose is in a low range. Concentration range and accuracy need are unknown, so achievable error cannot be judged.

medium Probe fouling and cell deposits (ATR crystal, windows, sapphire tips), bubbles and changing turbidity during a run can drift all signals. Fouling, temperature, pressure and sterilisation need are unknown.

low Hazardous area (ATEX) and laser safety for Raman are unknown. Fiber type, length and temperature rating of the existing link and probes are also unknown; MIR fiber links are much shorter than silica links.

medium The matrix, phase and water content are assumed (aqueous broth, liquid), not confirmed by the customer. Reference method, sampling plan and acceptance criteria are not yet defined, so the plan stays provisional.

## First feasibility experiment

### Test on real broth whether glucose and viable biomass can be measured with fiber probes, and check the premise that the existing fluorescence spectrometer can do this on sensitivity alone.

Samples

About 40–60 samples from at least 4 fermentation batches: 2 calibration batches plus 2 independent validation batches (final number to be agreed). — First check cell-free medium and real broth for fluorescence background and clarity. Then break the correlations of the process: (a) spike glucose into cell-free medium and into broth at constant cell density; (b) vary cell density at constant glucose by dilution or concentration; (c) mix viable and heat-killed cells at constant total biomass to separate viability from total biomass; (d) cover the expected temperature range and turbidity range, and different media or recipes if they exist.

Reference method

Customer's own methods; if none exist: enzymatic glucose assay or HPLC for glucose, and plate count or flow cytometry with viability staining for viable biomass (optical density only as a total biomass trend).

Methods compared

Raman, NIR, MIR, Fluorescence

Interferents

Medium and broth fluorescence (colour, vitamins, product), Total vs viable cells (dead cells, debris), Glucose consumption correlated with biomass growth, Temperature, Turbidity and bubbles (aeration, stirring), Other sugars and metabolites (e.g. ethanol, lactate, organic acids)

Duration

About 4–6 weeks for the lab study including sample preparation and reference analysis; an in-line fermentation campaign afterwards adds several weeks and depends on process access and GMP

rules.

1. Step 1: On cell-free medium and real broth samples, record Raman spectra (probe or window) and check background fluorescence, colour and turbidity; compare with the existing fluorescence spectrometer.
2. Step 2: Prepare the designed static samples (glucose spikes, cell density series, viable/heat-killed mixtures, temperature and turbidity variation) and take reference values with the agreed reference methods.
3. Step 3: Measure all samples with Raman (immersion probe) and NIR (transflection probe) fiber probes; add an MIR ATR probe if the first results show a gap. Measure the existing fluorescence spectrometer as a check of the customer's premise only, not as a candidate.
4. Step 4: Build calibrations on the two calibration batches, including single-method baselines for each method, and validate on the independent validation batches (split by batch, not by neighbouring spectra).
5. Step 5: Only if the lab results meet the criterion, plan an in-line run on one fermentation with paired reference samples at the probe position.
6. Step 6: Review with the customer: go/no-go per target, and whether glucose and viable biomass need different methods.

**Success criterion:** On independent validation batches, error (RMSEP) and mean bias for each target stay within limits agreed with the customer before the test, with bias not significantly different from zero. A model counts only if it is better than the single-method baselines and still works when glucose and biomass are decoupled by the designed samples.

## Parameters and attributes · 38 Facts

### Facts

Value · unit · status. Hover a row for the customer's words. Edit → Recalculate.

| Fact | Value | Unit | Status |
|---|---|---|---|
| MeasurementTask · measurement goal | quantify | | reported „…“ |
| MeasurementTask · target | glucose, viable_biomass | | reported „…“ |
| MeasurementTask · target definition | defined | | reported „…“ |
| MeasurementTask · spatial scope | bulk | | inferred |
| MeasurementTask · sampling plan | — | | unknown |
| MeasurementTask · reference method | — | | unknown |

| Fact | Value | Unit | Status |
|---|---|---|---|
| MeasurementTask · performance requirements | — | | unknown |
| MeasurementTask · time requirements | — | | unknown |
| MeasurementTask · target range | — | | unknown |
| MeasurementTask · business goal | — | | unknown |
| Sample · matrix | aqueous_bioprocess | | inferred |
| Sample · phase | liquid | | inferred |
| Sample · physical form | suspension | | inferred |
| Sample · optical properties · opacity | — | | unknown |
| Sample · optical properties · fluorescence | — | | unknown |
| Sample · heterogeneity | — | | unknown |
| Sample · variability · formulation | — | | unknown |
| Sample · variability · physical | — | | unknown |
| Sample · sample state constraints · contact | — | | unknown |
| Sample · sample state constraints · sampling | — | | unknown |
| Sample · relevant components · fluorophore | none_known | | reported „…“ |
| Sample · relevant components · absorber | — | | unknown |
| ProcessContext · deployment preference | in_line | | reported „…“ |
| ProcessContext · temperature range | — | | unknown |
| ProcessContext · temperature c | — | degC | unknown |
| ProcessContext · operation constraints · fouling | — | | unknown |

| Fact | Value | Unit | Status |
| --- | --- | --- | --- |
| ProcessContext · operation constraints · hazardous | — | | unknown |
| ProcessContext · instrument distance | — | | unknown |
| ProcessContext · assets · fiber | — | | unknown |
| ProcessContext · assets · fiber profile | — | | unknown |
| ProcessContext · assets · must reuse | — | | unknown |
| ProcessContext · assets · calibration scope | — | | unknown |
| ProcessContext · assets · method | Fluorescence | | reported „…“ |
| MeasurementTask · reference method · independent | — | | unknown |
| CustomerRequest · justification | sensitivity_only | | reported „…“ |
| CustomerRequest · proxy allowed | — | | unknown |
| CustomerRequest · asks | assessment | | reported „…“ |
| CustomerRequest · combination to keep calibration | — | | unknown |